WORKED SOLUTION

STEP II 1988, Question 5

Pure • Roots of Unity and Tangents

STEP II 1988, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q5

The key idea

Write $z = \cos\theta + \mathrm{i}\sin\theta$ and expand z^7 with the binomial theorem. The imaginary part is $\sin 7\theta$, and dividing it by $\cos^7\theta$ turns it into a polynomial in $t = \tan\theta$. The seventh roots of unity make $\sin 7\theta$ vanish, so they hand us seven roots of a degree-7 polynomial for free. After that the tangent product is just the product of the roots, read off from the constant term.

Solution

The roots of the sextic

By de Moivre's theorem, $(\cos\theta + \mathrm{i}\sin\theta)^7 = \cos 7\theta + \mathrm{i}\sin 7\theta$. Write $c = \cos\theta$, $s = \sin\theta$. Expanding the left-hand side with the binomial theorem and keeping the terms with an odd power of is gives

$$\sin 7\theta = 7c^6 s - 35c^4 s^3 + 21c^2 s^5 - s^7.$$

Whenever $\cos\theta \neq 0$ we may divide by c^7 and put $t = \tan\theta$:

$$\frac{\sin 7\theta}{\cos^7\theta} = 7t - 35t^3 + 21t^5 - t^7 = -t\,(t^6 - 21t^4 + 35t^2 - 7). \qquad (*)$$

The roots of $z^7 = 1$ are $z = \cos\theta + \mathrm{i}\sin\theta$ with $\theta = \frac{2k\pi}{7}$, and for these $\sin 7\theta = \sin 2k\pi = 0$, the imaginary part of $z^7 = 1$. It is a little tidier to use $\theta = \frac{k\pi}{7}$, $k = 0, 1, \ldots, 6$, which also gives $\sin 7\theta = \sin k\pi = 0$ (and gives the same set of tangents, because tan has period π).

For each of these θ:

- $\cos\theta \neq 0$, since $\frac{k\pi}{7} = \frac{\pi}{2}$ would need $2k = 7$; so $(*)$ is valid;
- $\sin 7\theta = 0$, so by $(*)$ the number $t = \tan\frac{k\pi}{7}$ is a root of $p(t) = t^7 - 21t^5 + 35t^3 - 7t$.

The seven values $\tan\frac{k\pi}{7}$ ($k = 0, \ldots, 6$) are distinct: the angles $0, \frac{\pi}{7}, \ldots, \frac{6\pi}{7}$ are distinct points of $[0, \pi)$, none equal to $\frac{\pi}{2}$, and tan is one-to-one on $[0, \pi) \setminus \{\frac{\pi}{2}\}$ (positive and increasing on $[0, \frac{\pi}{2})$, negative on $(\frac{\pi}{2}, \pi)$, also increasing there, with the two pieces taking values in $[0, \infty)$ and $(-\infty, 0)$).

Ask yourself How do we know we have found *all* the roots, not just some?

A polynomial of degree 7 has at most 7 roots. We have found 7 distinct ones, so they are all of them. The value $k = 0$ gives $t = 0$, the root coming from the factor t. Hence the roots of $t^6 - 21t^4 + 35t^2 - 7 = 0$ are exactly

$$t = \tan\frac{k\pi}{7}, \qquad k = 1, 2, 3, 4, 5, 6,$$

that is, $\pm\tan\frac{\pi}{7}$, $\pm\tan\frac{2\pi}{7}$, $\pm\tan\frac{3\pi}{7}$, since $\tan\frac{(7-k)\pi}{7} = -\tan\frac{k\pi}{7}$.

The product $\tan\frac{2\pi}{7}\tan\frac{4\pi}{7}\tan\frac{6\pi}{7}$

For a monic polynomial of even degree, the product of the roots equals the constant term. So

$$\left(\tan\tfrac{\pi}{7}\right)\left(-\tan\tfrac{\pi}{7}\right)\left(\tan\tfrac{2\pi}{7}\right)\left(-\tan\tfrac{2\pi}{7}\right)\left(\tan\tfrac{3\pi}{7}\right)\left(-\tan\tfrac{3\pi}{7}\right) = -7,$$

which says

$$\left(\tan\tfrac{\pi}{7}\tan\tfrac{2\pi}{7}\tan\tfrac{3\pi}{7}\right)^2 = 7.$$

All three angles lie in $(0, \frac{\pi}{2})$, so each tangent is positive, and the product is the positive square root: $\tan\frac{\pi}{7}\tan\frac{2\pi}{7}\tan\frac{3\pi}{7} = \sqrt{7}$.

Now use $\tan(\pi - x) = -\tan x$: $\tan\frac{4\pi}{7} = -\tan\frac{3\pi}{7}$ and $\tan\frac{6\pi}{7} = -\tan\frac{\pi}{7}$. The two minus signs cancel, so

$$\tan\frac{2\pi}{7}\tan\frac{4\pi}{7}\tan\frac{6\pi}{7} = \tan\frac{2\pi}{7}\tan\frac{3\pi}{7}\tan\frac{\pi}{7} = \sqrt{7},$$

as required.

The cases $n = 9$ and $n = 11$

The same argument works for any odd $n = 2m + 1$. Taking the imaginary part of $(\cos\theta + \mathrm{i}\sin\theta)^n$ and dividing by $\cos^n\theta$,

$$\frac{\sin n\theta}{\cos^n\theta} = \sum_{r=0}^{m}(-1)^r\binom{n}{2r+1}t^{2r+1} = t\,q_n(t),$$

where q_n has degree $2m = n - 1$, leading coefficient $(-1)^m$ and constant term $\binom{n}{1} = n$. Explicitly,

$$q_9(t) = t^8 - 36t^6 + 126t^4 - 84t^2 + 9,$$
$$q_{11}(t) = -t^{10} + 55t^8 - 330t^6 + 462t^4 - 165t^2 + 11.$$

Exactly as before (n is odd, so no $\frac{k\pi}{n}$ equals $\frac{\pi}{2}$), the roots of q_n are the $n-1$ distinct numbers $\tan\frac{k\pi}{n}$, $k=1,\dots,n-1$, which pair up as $\pm\tan\frac{k\pi}{n}$ for $k=1,\dots,m$. The product of the roots of q_n is (constant term)/(leading coefficient), because the degree is even:

$$(-1)^m\prod_{k=1}^{m}\tan^2\frac{k\pi}{n}=\frac{n}{(-1)^m}\quad\Longrightarrow\quad\prod_{k=1}^{m}\tan\frac{k\pi}{n}=\sqrt{n},$$

taking the positive root since all these angles are in $(0,\frac{\pi}{2})$. For $n=9$ this is q_9's constant term 9 over leading coefficient 1; for $n=11$ it is $11/(-1)=-11=(-1)^5\prod\tan^2$. Either way, the product of $\tan\frac{k\pi}{n}$ over $k=1,\dots,m$ is 3 for $n=9$ and $\sqrt{11}$ for $n=11$.

Ask yourself The required product has angles $\frac{2\pi}{n},\frac{4\pi}{n},\dots$, not $\frac{\pi}{n},\frac{2\pi}{n},\dots$. How do we get from one to the other, and what happens to the sign?

Each even multiple $2j\le m$ is already in the list $1,\dots,m$. Each even $2j>m$ (that is, $2j>\frac{n}{2}$) gives $\tan\frac{2j\pi}{n}=-\tan\frac{(n-2j)\pi}{n}$, and $n-2j$ is odd and at most m. Every number from 1 to m arises exactly once in this way, because the even ones come from the first kind and the odd ones from the second. So the required product is $\pm\sqrt{n}$, with one minus sign for each even number between $\frac{n}{2}$ and n.

- $n=9$: the even numbers $2,4,6,8$; those above 4.5 are $6,8$, two minus signs. So
$$\tan\frac{2\pi}{9}\tan\frac{4\pi}{9}\tan\frac{6\pi}{9}\tan\frac{8\pi}{9}=\sqrt{9}=3.$$
- $n=11$: the even numbers $2,4,6,8,10$; those above 5.5 are $6,8,10$, three minus signs. So
$$\tan\frac{2\pi}{11}\tan\frac{4\pi}{11}\tan\frac{6\pi}{11}\tan\frac{8\pi}{11}\tan\frac{10\pi}{11}=-\sqrt{11}.$$

(For $n=7$ the even numbers above 3.5 are $4,6$, two signs, which is why that product came out positive.)

Checks

- $n=9$ has an easy partial check: $\tan\frac{6\pi}{9}=\tan\frac{2\pi}{3}=-\sqrt{3}$, and numerically $\tan\frac{2\pi}{9}\tan\frac{4\pi}{9}\tan\frac{8\pi}{9}\approx 0.8391\times 5.6713\times(-0.3640)\approx -1.732=-\sqrt{3}$, giving a total of 3.
- On a calculator, $\tan\frac{2\pi}{7}\tan\frac{4\pi}{7}\tan\frac{6\pi}{7}\approx 2.645751$ and $\sqrt{7}\approx 2.645751$; the $n=11$ product is $\approx -3.316625=-\sqrt{11}$.
- $\tan\frac{\pi}{7}\approx 0.4816$ gives $t^6-21t^4+35t^2-7\approx 0$, as it should.

Common slips

- Dividing by $\cos^7\theta$ without checking that it is non-zero, or not arguing that the seven tangents are distinct. Without these points you have not shown that you have found

all the roots.

- Getting the sign wrong for $n = 11$: the square of the product is easy, but the sign needs the count of negative tangents.
- Forgetting that the product of the roots of $a_d t^d + \cdots + a_0$ is $(-1)^d a_0/a_d$, which matters for $n = 11$, where the leading coefficient is -1.

Beyond STEP

The numbers $\tan \frac{k\pi}{n}$ are algebraic, and the polynomials q_n are their minimal polynomials (when n is prime) over the rationals. Identities like $\prod \tan \frac{k\pi}{n} = \sqrt{n}$ are shadows of Gauss sums, which show that $\sqrt{\pm p}$ lies in the field generated by the pth roots of unity, a cornerstone of algebraic number theory. Davenport's *The Higher Arithmetic* is a gentle way into this circle of ideas, and Needham's *Visual Complex Analysis* gives a geometric feel for roots of unity.

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